ResearchOS: Closed-Loop, Evidence-Grounded Research Architecture

Human-guided orchestration, adaptive retrieval, specialized scientific agents, and claim-level provenance

HUMAN-GUIDED CONTROL PLANE

1. **Research Goal + Project Context** — prompt, files, history, constraints
2. **Context Compiler** — tenant scope and working memory
3. **Dynamic Planner + Task Scheduler** — workflow graph, dependencies, retry, resume
4. **Agent + Tool Router** — models, search, code, MCP tools
5. **Run Monitor + Checkpoints** — approval, status, audit, cancellation

EVIDENCE FOUNDATION

Private and scholarly knowledge

- **Project Library** — PDF, DOCX, text, data
- **Scholarly Search** — papers and metadata
- **Document Intelligence** — parse - normalize - semantic chunk - embed
- **Hybrid Retrieval** — dense + lexical search - reciprocal-rank fusion; project and tenant scoped
- **Traceable Evidence Pack** — ranked passages + source/page/chunk IDs

relevance · diversity · provenance

MULTI-AGENT SCIENTIFIC REASONING LOOP

Discover, propose, evaluate, and revise

unsupported or contradicted claims trigger retrieval, revision, or replanning

1. **Discover**
   - Literature Agent
   - Evidence Mapping
   - Gap Analysis

   search, deduplicate, synthesize

2. **Hypothesize**
   - Hypothesis Generator
   - Hypothesis Lineage
   - Candidate Tournament

   falsifiable, evidence-linked proposals

3. **Evaluate**
   - Novelty Auditor + Prior Art
   - Feasibility Agent
   - Code + Experiment Runner

   testability, risk, methods, results

4. **Critique + Synthesize**
   - Scientific Critic
   - Statistical Reviewer
   - Academic Writing Agent

   accept, revise, reject, or replan

Shared Research State — plans - observations - evidence; claims - artifacts - decisions

evidence on demand

TRUST AND DELIVERY

Only supported claims advance

- **Claim Verification** — support, contradiction, experiment provenance
- **Citation Validation** — claim-to-source integrity and sequence
- **Context-Aware Drafting** — document history + selected citation style; IEEE - APA 7 - Harvard - MLA 9
- **Research Deliverables** — draft + references + DOCX/PDF export

researcher review and revision

PERSISTENT DATA, MEMORY, AND GOVERNANCE PLANE

- **PostgreSQL Research State** — projects, runs, conversations
- **Document + Artifact Store** — sources, drafts, results, exports
- **Vector + Lexical Index** — embeddings and searchable chunks
- **Evidence + Lineage Graph** — claims, citations, ancestry
- **Model + Tool Gateways** — LLMs, search, code, MCP
- **Tenant Security + Audit** — authorization, approvals, usage

primary workflow · evidence flow · quality-control feedback · persistence and shared state

Modes: deep research - idea discovery - paper review - reproduction - research debugging